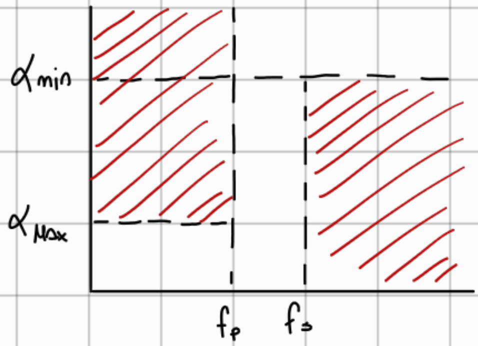
Plantilla.



α_{max} [db]	w_p	w_s
1	1	1,25

1) Queremos un retardo de grupo cte, mismo retardo para todas las frec.

$$D(w) = -\frac{\partial(\varphi(w))}{\partial w}$$

Aproximación Storch.

$$H(jw)\Big|_{w=s/j} = e^{\cancel{-j}\frac{st}{\cancel{j}}} = e^{-st}$$

$$H(jw) = \frac{P(s)}{q(s)} = e^{-st} = \frac{1}{\operatorname{senh}(s)+\cosh(s)}$$

$$\operatorname{senh}(x) = \frac{e^{x}-e^{-x}}{2} \qquad \cosh(x) = \frac{e^{x}+e^{-x}}{2}$$

$$e^{x} = \operatorname{senh}(x) + \cosh(x)$$

$m=1$

$$\operatorname{Cotg}(s) = \frac{1}{s} = \frac{\cosh(s)}{\operatorname{senh}(s)}$$

$$\operatorname{senh}(s) = s + \frac{s^3}{3!} + \frac{s^5}{5!} + \frac{s^7}{7!} + \dots$$

$$H(s) = \frac{1}{\operatorname{senh}(s)+\cosh(s)} = \frac{1}{s+1}$$

$$\cosh(s) = 1 + \frac{s^2}{2!} + \frac{s^4}{4!} + \frac{s^6}{6!} + \dots$$

$$\boxed{H_{B1}(s) = \frac{1}{s+1}}$$

expanción en fracciones continuas

$$\operatorname{Cotgh}(s) = \frac{\cosh(s)}{\operatorname{senh}(s)} = \frac{1}{s} + \cfrac{1}{\frac{3}{s} + \cfrac{1}{\frac{5}{s} + \cfrac{1}{\frac{7}{s} + \cfrac{1}{\frac{9}{s} + \dots}}}}$$

sabiendo que $D(w) = \frac{a_1}{a_0}\left[1 + \left(\frac{1}{a_0} - \frac{a_1^2}{a_0^2} + \frac{2}{a_0}\right)w^2 + \dots\right]$

Para $D(0) = \frac{a_1}{a_0} = \frac{1}{1} \to \boxed{D(0) = 1}$

todos los impares.

$m=2$

$$\operatorname{Cotgh}(s) = \frac{1}{s} + \frac{s}{3} = \frac{3+s^2}{3s}$$

$$\boxed{H_{B2}(s) = \frac{3}{s^2+3s+3}}$$

$$D(0) = \frac{a_1}{a_0} = \frac{3}{3} \longrightarrow \boxed{D(0) = 1}$$

$\underline{n=3}$

$$Cotgh(s) = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{s}{5}} = \frac{1}{s} + \frac{5s}{15+s^2} = \frac{6s^2+15}{s^3+(15)s}$$

$$\boxed{H_{B3}(s) = \frac{15}{s^3+6s^2+(15)s+15}} \qquad D(\phi) = \frac{a_1}{a_0} = \frac{15}{15} \longrightarrow \boxed{D(\phi)=1}$$

$\underline{n=4}$

$$Cotgh(s) = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{1}{\frac{5}{s} + \frac{1}{\frac{7}{s}}}} = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{1}{\frac{5}{s} + \frac{s}{7}}} = \frac{1}{s} + \frac{1}{\frac{3}{s} + \frac{(7)s}{35+s^2}}$$

$$Cotgh(s) = \frac{1}{s} + \frac{1}{\frac{105+(3)s^2+7s^2}{s^3+(35)s^2}} = \frac{1}{s} + \frac{s^3+(35)s}{(10)s^2+(105)} = \frac{s^4+(45)s^2+105}{(10)s^3+(105)s}$$

$$Cotgh(s) = \frac{s^4+(45)s^2+105}{(10)s^3+(105)s}$$

$$\boxed{H_{B4}(s) = \frac{105}{s^4+(10)s^3+(45)s^2+(105)s+105}} \qquad D(0) = \frac{a_1}{a_0} = \frac{105}{105}$$

$$\boxed{D(0)=1}$$

2) $$\alpha = -20\log(|H(j\omega)|) = 10\log(|H(j\omega)|^{-2}) = 10\log\left(\frac{B_n(j\omega_s)^2}{B_n(\phi)^2}\right)$$

$\underline{\text{P}_{\text{ara}}\ n=2}$

$$H_{B2}(j\omega) = \frac{3}{(j\omega)^2+(j\omega)(3)+3} \longrightarrow \omega_p=1 \quad H(j) = \frac{3}{-1+j3+3}$$

$$|H(j)|^2 = \frac{(3)^2}{(2)^2+(3)^2} \longrightarrow \alpha(1r/s) = 10\log\left(\frac{(2)^2+(3)^2}{(3)^2}\right) \longrightarrow \boxed{\alpha(1r/s) = 1{,}59} > \alpha_{max} \quad \text{X}$$

$\underline{\text{P}_{\text{ara}}\ n=3}$

$$H_{B3}(j\omega) = \frac{15}{(j\omega)^3+6(j\omega)^2+15(j\omega)+15} \longrightarrow H_{B3}(j) = \frac{15}{-j-6+j15+15} = \frac{15}{9+j14}$$

$$|H_{B3}(j)|^2 = \frac{196}{81+196} \qquad \alpha(1r/s) = 10\log\left(\frac{277}{225}\right) \longrightarrow \boxed{\alpha(r/s) = 0{,}9} \quad \text{V}$$

La $T(s)$ que cumple con el α_{max} es la de orden $N=3$

$$T_{B3}(s) = \frac{15}{s^3 + s^2 6 + 15s + 15}$$

3)

$$D_{B3}(\omega) = \frac{6\omega^4+45\omega^2+225}{\omega^6+6\omega^4+45\omega^2+225} \rightarrow D_{B3}(2,5) = \frac{6(2,5)^4+45(2,5)^2+225}{(2,5)^6+6(2,5)^4+45(2,5)^2+225}$$

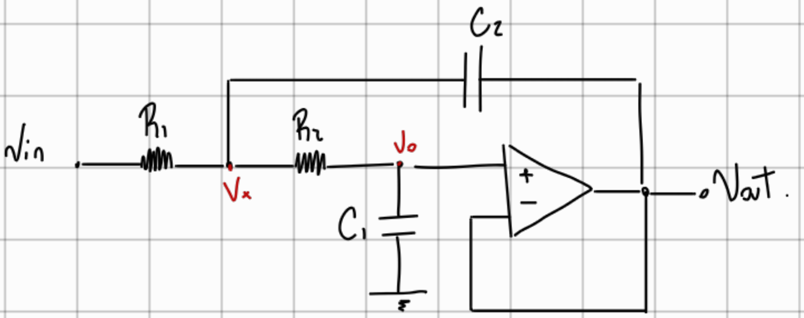
$$\boxed{D_{B3}(2,5) = 0,752 \text{ seg.}}$$

$$\text{Error} = \frac{D(2,5)-D(0)}{D(0)}\cdot 100$$

$$\text{Error} = \frac{0,752-1}{1}\cdot 100 \Rightarrow \boxed{\text{Error} = -24,8\%}$$

4)

Topología Sallen-Key



Con $k=1$

(1) $V_x(G_1+G_2+sC_2) = V_o(G_2+sC_2)+V_iG_1$

(2) $V_o = V_x \dfrac{G_2}{sC_1+G_2}$

$$V_x = V_o\frac{sC_1+G_2}{G_2}$$

$$V_o\left[\frac{(sC_1+G_2)(G_1+G_2)+s^2C_1C_2+sC_1G_2}{G_2} - G_2 - sC_2\right] = V_iG_1$$

$$V_o\left[\frac{sC_1G_1+sC_1G_2+G_1G_2+\cancel{G_2^2}+s^2C_1C_2+\cancel{sC_2G_2}-\cancel{G_2^2}-\cancel{sC_2G_2}}{G_1G_2}\right]=V_i$$

$$V_o\left[\frac{s^2C_1C_2+sC_1G_2+sC_1G_1+G_1G_2}{G_1G_2}\right]=V_i \rightarrow \frac{V_o}{V_i} = \frac{G_1G_2}{C_1C_2}\cdot\frac{1}{s^2+s\frac{1}{C_2}(G_1+G_2)+\frac{G_1G_2}{C_1C_2}}$$

$$T_{SK}(s) = \frac{V_o(s)}{V_i(s)} = \frac{G_1G_2}{C_1C_2}\cdot\frac{1}{s^2+s\frac{1}{C_2}(G_1+G_2)+\frac{G_1G_2}{C_1C_2}} = \frac{\overbrace{k}^{=1}\cdot\omega_0^2}{s^2+s\frac{\omega_0}{Q}+\omega_0^2}$$

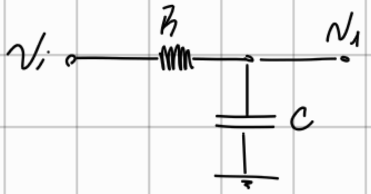
$$\boxed{T_{SK}(s) = \frac{\frac{G_1G_2}{C_1C_2}}{s^2+s\frac{1}{C_2}(G_1+G_2)+\frac{G_1G_2}{C_1C_2}}}$$

factorizo $H_{B3}(s) = \dfrac{15}{s^3 + 6s^2 + (15)s + 15} \longrightarrow = \emptyset \quad s = -2,322$

$-2,322$	1	6	15	15
		$-2,322$	$-8,54$	-15
	1	3,678	6,46	$\boxed{\emptyset}$

$$H_{B3}(s) = \frac{1}{(s+2,322)} \cdot \frac{15}{\left(s^2 + s(3,678) + 6,46\right)}$$

El polinomio de Bessel es de orden 3 por lo tanto utilizo dos circuitos en Cascada



$$t_1(s) = \frac{V_1}{V_i} = \frac{1}{C} \cdot \frac{1}{s + 1/CR}$$

$$t(s) = t_1(s) \cdot t_{s-k}(s) = \left(\frac{1}{C_3}\right)\left(\frac{1}{s + \frac{1}{C_3 R_3}}\right)\left(\frac{G_1 G_2}{C_1 C_2}\right)\left(\frac{1}{s^2 + s\frac{1}{C_2}(G_1 + G_2) + \frac{G_1 G_2}{C_1 C_2}}\right)$$

Normalización

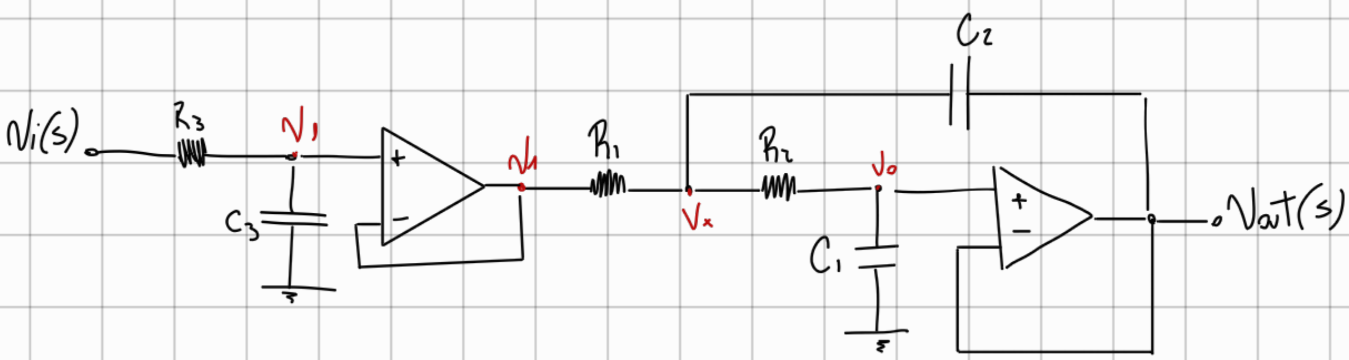
$R_3 = 1 \qquad \dfrac{1}{C_3 \cdot R_3} = 2,322 \longrightarrow C_3 = \dfrac{1}{(2,322)R_3} \qquad G_1 = G_2 = 1 \qquad \dfrac{1}{C_1 C_2} = 6,46$

$\dfrac{w_0}{q} = 3,678 = \dfrac{1}{C_2}(G_1 + G_2) \qquad C_1 = \dfrac{1}{C_2(6,46)}$

$C_2 = \dfrac{2G_1}{3,678}$

$G_1 = G_2 = G_3 = 1$

$C_1 = \dfrac{1}{C_2 \cdot 6,46} \qquad C_2 = \dfrac{2G_1}{3,678} \qquad C_3 = \dfrac{1}{(2,322)R_3}$



Desnormalización para $D_0(0) = 200\ \mu s$.

$s = \dfrac{s'}{\frac{1}{200\mu s}} \longrightarrow s = \dfrac{s}{5000}$ // Mantengo $G_1 = G_2 = G_3 = 1$ entonces:

Considerando la impedancia del cap. $Z_c = \dfrac{1}{sC} \longrightarrow Z_c = \dfrac{1}{s\,\frac{C}{5000}}$

Desnormalizo

$C_3' = \dfrac{C_3}{5000}$ $\quad C_3 = \dfrac{1}{2{,}322} = 0{,}43$ $\quad \boxed{C_3' = 86\ \mu F}$

$C_2' = \dfrac{C_2}{5000}$ $\quad C_2 = \dfrac{2}{3{,}678} = 0{,}5437$ $\quad \boxed{C_2' = 108{,}74\ \mu F}$

$C_1' = \dfrac{C_1}{5000}$ $\quad C_1 = \dfrac{1}{C_2(6{,}46)} = 0{,}2846$ $\quad \boxed{C_1' = 56{,}92\ \mu F}$